Eenheid 2.2

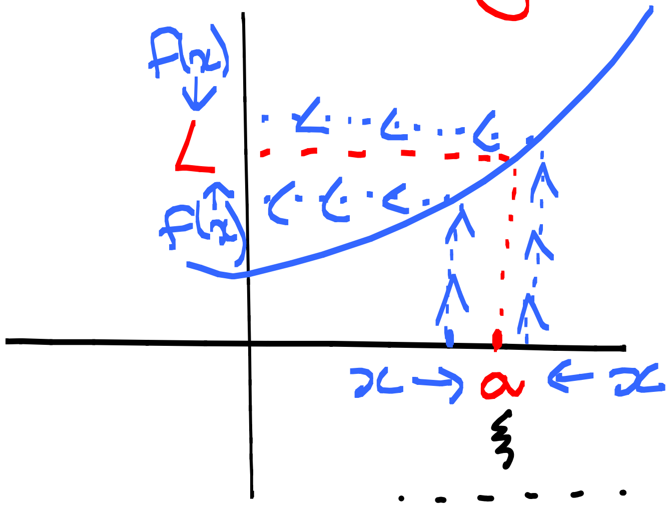
Note Title 2016/02/24

NB Die limiet van 'n funksie

Stewart bl. 83-94
Studiegids bl. 13

Definisie (informeel)

Veronderstel $f(x)$ is gedefinieer as x naby die getal a is.



dws. f is gedefinieer op 'n oop interval wat a bevat, maar $x \neq a$.

"Soos wat x-waardes nader kom na a maar nooit self a is nie, kom f(a) nader na L."

Ons skryf dit as volg:

$$\lim_{x \to a} f(x) = L$$

Kyk bv. na • $f(x) = \frac{e^{4x}-1}{x}$ $f\left(\frac{2}{3}\right) = \frac{e^{8}-1}{2}$

≈ 1489.978

• $f(0) = \frac{e^{0}-1}{0}$ ongedefinieerd

Wat gebeur met die funksie $f(x) = \frac{e^{4x}-1}{x}$ as ons na waardes naby 0 kyk?

x positief	$f(x)$
0.01	4.081077
0.001	4.0080167

x negatief	$f(x)$
−0.01	3.921056
−0.001	3.992010656

Hoe nader die waardes na 0 kom, van beide kante af (maar $x \neq 0$) hoe nader kom die waardes van $f(x)$ na 4. • Dus $\lim_{x \to 0} \frac{e^{4x}-1}{x} = 4$

VOORBEELDE: Tegnieke om limiete te bepaal.

Probeer altyd eers die volgende metode

*Direkte substitusie

1. $$\lim_{x \to -1} (x^2 + x - 2) = (-1)^2 + (-1) - 2 = -2$$

2. $$\lim_{x \to 2} \frac{x^2 - 4}{x + 2} = \frac{4 - 4}{4} = 0$$

As direkte substitusie nie werk nie

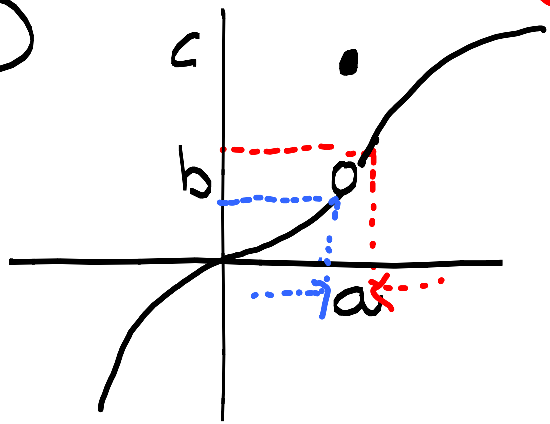
"Plan": Faktoriseer

1. $$\lim_{x \to -3} \frac{x^2 - 9}{x + 3} \quad \left(\frac{0}{0}\right) \xrightarrow{\text{onbepaalde vorm}} = \lim_{x \to -3} \frac{\cancel{(x+3)}(x-3)}{\cancel{x+3}} = \lim_{x \to -3} x - 3 = -6$$

Eensydige limiete (halwe limiete)

Stuksgewys gedefineerde funksies.

①



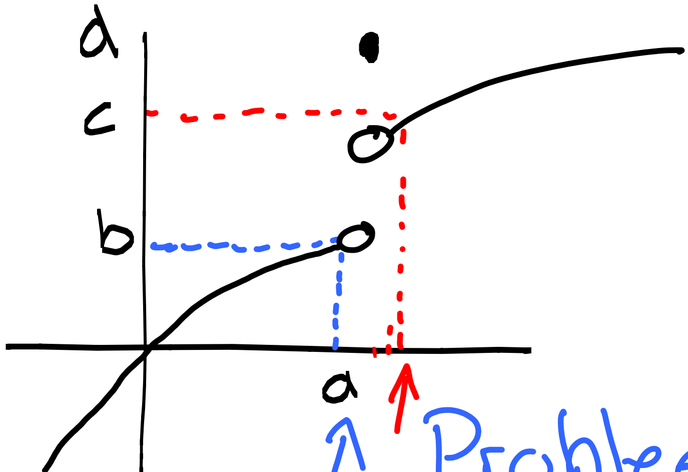
- $f(a) = c$
- $\lim\limits_{x \to a^+} f(x) = \boxed{b}$ (limiet as x van regs af al hoe nader aan a kom)

  Regterlimiet
- $\lim\limits_{x \to a^-} f(x) = \boxed{b}$ (x kom al hoe nader na a van links af)

  Linkerlimiet

Dan $\lim\limits_{x \to a} f(x) = b$

②



- $\lim\limits_{x \to a^+} f(x) = c$ (regterlimiet)
- $\lim\limits_{x \to a^-} f(x) = b$ (linkerlimiet)

Probleem: $\lim\limits_{x \to a^+} f(x) \neq \lim\limits_{x \to a^-} f(x)$

Stelling:

$$\lim_{x\to a} f(x) = L \iff \lim_{x\to a^+} f(x) = L \text{ en } \lim_{x\to a^-} f(x) = L$$

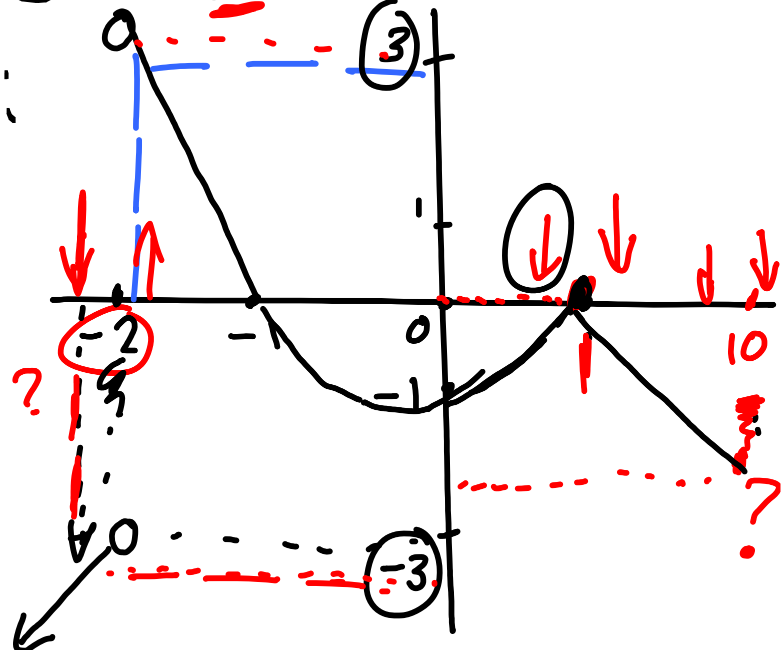
(bestaan)

Voorbeelde

① Gegee $f(x) = \begin{cases} 2x+1 & \text{as } x < -2 \\ x^2-1 & \text{as } -2 < x \le 1 \\ 1-x & \text{as } x > 1 \end{cases}$ ($x \to 10$)

Vind, indien moontlik:

- $\lim_{x\to 10} f(x)$ ✓
- $\lim_{x\to 1} f(x) = 0$ ✓
- $\lim_{x\to -2} f(x)$?



- $\lim\limits_{x \to 10} f(x)$ (as $x \to 10$ is $f(x) = 1-x$)

$= \lim\limits_{x \to 10} (1-x) = -9$

- $\lim\limits_{x \to 1} f(x) = ?$

$\lim\limits_{x \to 1^+} f(x) = \lim\limits_{x \to 1^+} (1-x)$ ($x > 1$ en $f(x) = 1-x$)

$= 0$

$\lim\limits_{x \to 1^-} f(x) = \lim\limits_{x \to 1^-} (x^2-1) = 0$ ($x < 1$ en $f(x) = x^2-1$)

Dus $\lim\limits_{x \to 1} f(x) = 0$ ✓ want $\lim\limits_{x \to 1^-} f(x) = \lim\limits_{x \to 1^+} f(x) = 0$

• $\lim_{x\to -2} f(x) = ?$

$$\lim_{x\to -2^-} f(x) = \lim_{x\to -2^-} (2x+1) \qquad (x<-2)$$

$$= -4+1 = -3$$

$$\lim_{x\to -2^+} f(x) = \lim_{x\to -2^+} (x^2-1) = 4-1=3 \qquad (x>-2)$$

Dus $\lim_{x\to -2} f(x)$ bestaan nie want $\lim_{x\to -2^-} f(x) \neq \lim_{x\to -2^+} f(x)$

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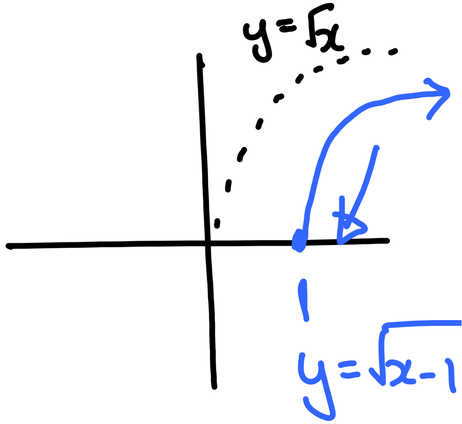
② $\lim_{x\to 1} \sqrt{x-1}$?

$$\lim_{x\to 1^+} \sqrt{x-1} = 0$$



***Redenasie:**

$x \to 1^+ \Rightarrow x>1 \Rightarrow x-1>0$ ✓

* **Redenasie:** $\lim_{x\to1^-}\sqrt{x-1}$?? $\quad x\to1^- \Rightarrow x<1 \Rightarrow x-1<0$

ongedefinieerd

Gevolgtrekking:

$\lim_{x\to1}\sqrt{x-1}$ bestaan nie want $\lim_{x\to1^-}\sqrt{x-1}$ bestaan nie

maar $\lim_{x\to1^+}\sqrt{x-1} = 0$

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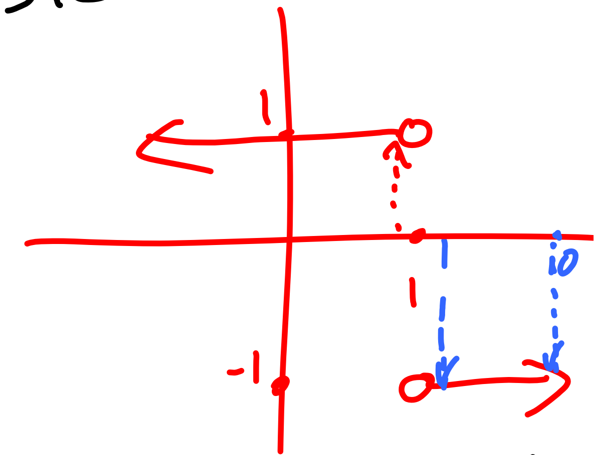
③ $f(x) = \dfrac{1-x}{|x-1|}$ $\quad$ • $\lim_{x\to1} f(x) = ?$ $\quad$ • $\lim_{x\to10} f(x) = ?$

$$= \begin{cases} \dfrac{1-x}{(x-1)} & \text{as } x-1>0 \\[2ex] \dfrac{1-x}{-(x-1)} & \text{as } x-1<0 \end{cases}$$



$$= \begin{cases} \dfrac{1-x}{-(1-x)} & \text{as } x>1 \end{cases} = \begin{cases} -1 & \text{as } x>1 \\ 1 & \text{as } x<1 \end{cases}$$

$$\begin{cases} \dfrac{1-x}{1-x} & \text{as } x < 1 \end{cases}$$

$\lim_{x \to 1} f(x) = ??$ bestaan nie Want $\lim_{x \to 1^-} f(x) = 1 \neq \lim_{x \to 1^+} f(x) = -1$

$\lim_{x \to 10} f(x) = -1$

• Oneindige limiete

(antwoord is $\pm\infty$)

Informele definisie:

$\lim_{x \to a} f(x) = \infty$ Waardes van $f(x)$ kan willekeurig groot gemaak word (so groot as wat jy wil) deur x naby genoeg aan a te neem, maar $x \neq a$

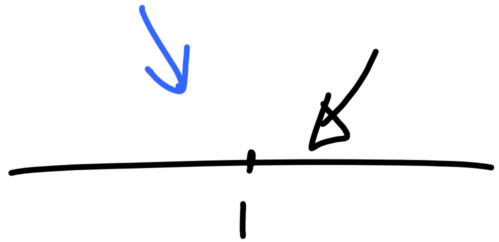
Dus $\lim_{x \to a} f(x) = \infty$ beteken

Dus limiet bestaan nie !! Limiet bestaan net as limiet se antwoord 'n reële getal is.

Bv. • $\lim_{x\to 1} x+2 = 3$

$$\lim_{x\to 1} \frac{1 \; (\neq 0)}{(x-1)^2} = \infty$$

0.9   1.1   word al kleiner



Een tipe oneindige limiete is in 'n vorm wat jy sal moet kan herken.

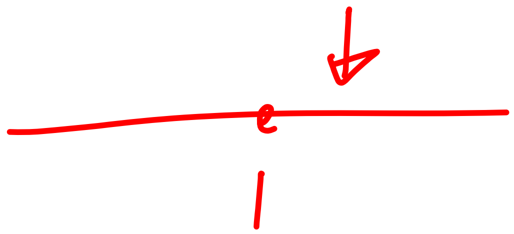
$$\lim_{x\to a}\left[\frac{\text{nie-nul getal}}{\text{getal wat kleiner en kleiner word}}\right] = \pm\infty\,?$$

Bv $\lim_{x\to 1} \dfrac{1 \; (\neq 0)}{(x-1)^2} = \boxed{+}\,\infty$? antw al hoe groter

kleiner

$\left(\frac{-}{+}\right)$

- $\lim\limits_{x \to 1^+} \dfrac{\overset{\neq 0}{-4}}{(x-1)_{+}} = -\infty$



$x \to 1^+ \Rightarrow x > 1$

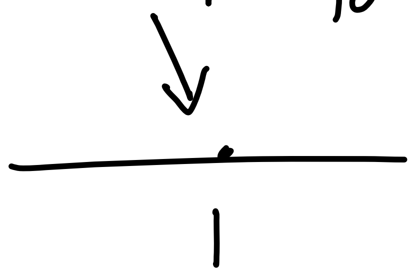
$\Rightarrow x - 1 > 0$ en $-4 < 0$

- $\lim\limits_{x \to 1^-} \dfrac{\overset{\neq 0}{-4}}{x-1} = \boxed{+}\infty$?

groter

$x - 1 \rightsquigarrow$ kleiner

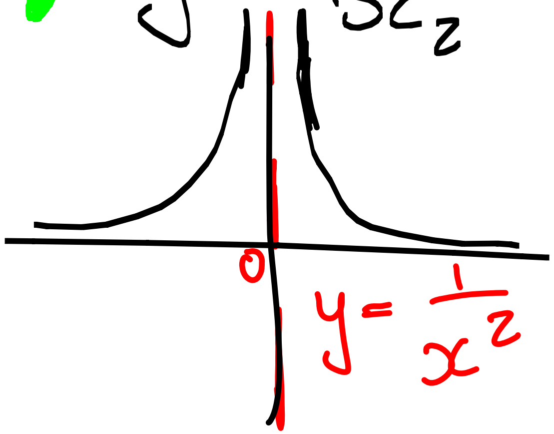
$0.9 = \frac{9}{10}$



*Vertikale asimptote

Spesiale vbd.

• $y = \frac{1}{x^2}$



STAPPE:

• $D_f = (-\infty, 0) \cup (0, \infty) = \{x \mid x \neq 0\}$

*Is $0 \in D_f$? Nee!

• Vermoed daar is 'n Vertikale asimptoot by $x = 0$.

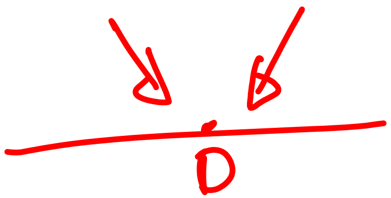
NB • Moet dit toets/bevestig!!!

NB *Toets:

$\lim_{x \to 0^-} \frac{1}{x^2} = +\infty$

$\lim_{x \to 0^+} \frac{1}{x^2} = +\infty$

VA by $\boxed{x = 0}$



Teorie:

$x = a$ is 'n vertikale asimptoot van $y = f(x)$ as ten minste een van die volgende bewerings geld:

$$\lim_{x \to a} f(x) = +\infty \qquad \lim_{x \to a} f(x) = -\infty$$

$$\lim_{x \to a^+} f(x) = +\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

$$\lim_{x \to a^-} f(x) = +\infty \qquad \lim_{x \to a^-} f(x) = -\infty$$

Gevolgtrekking: (Vir gewone funksies!)

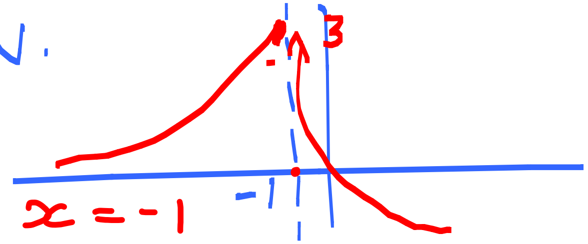
- Bepaal D_f.

As $x=a$ in D_f is, dan is $x=a$ nie 'n vertikale asimptoot nie

As $x=a$ nie in D_f is nie, dan kan $x=a$ MOONTLIK 'n VA wees.

- Toets of $x=a$ 'n VA is of nie!!

Maar • Stuksgewysgedefineerde funksies kan probleme skep! Bv.



Plan: $-1 \in D_f$ maar $\lim_{x \to -1^+} f(x) = \infty$ $\therefore$ VA is $x=-1$

VOORBEELDE:

① $f(x) = x+1$

$D_f = \mathbb{R}$ Geen VA

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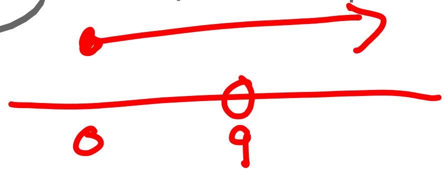
② $f(x) = \dfrac{\sqrt{\underset{\geq 0}{x}} - 3}{x-9}$ $\rightarrow x \neq 9$ $D_f = [0,9) \cup (9,\infty)$



Vermoed VA by $x = 9$

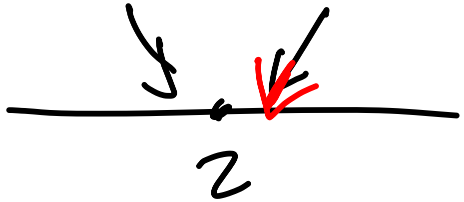
- Toets: $\lim\limits_{x \to 9} \dfrac{\sqrt{x} - 3}{x-9} \left(\dfrac{0}{0}\right)$

$= \lim\limits_{x \to 9} \dfrac{\sqrt{x} \cancel{- 3}}{(\sqrt{x} \cancel{-3})(\sqrt{x}+3)} = \boxed{\dfrac{1}{6}}$

Dus geen VA

- Metode: limiete
  - Direk
  - ✓ $\left(\frac{0}{0}\right)$ Faktoriseer
  - Halwe limiete

③ $$\lim_{x \to 2} \frac{1}{(x-2)^2} = {}^{+}\infty$$



Dus VA is by $\underline{\underline{x = 2}}$

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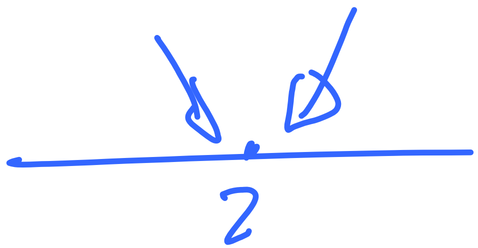
④ $$\lim_{x \to 2} \frac{1}{x-2} = ?$$



$$\lim_{x \to 2^+} \frac{1}{(x-2)_+} = {}^{+}\infty$$

$$\lim_{x \to 2^-} \frac{1}{(x-2)_-} = -\infty$$

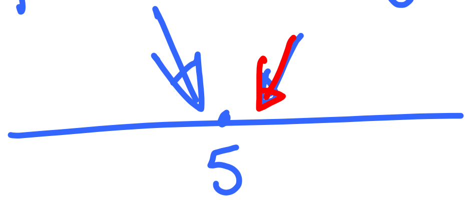
Dus VA by $x = 2$

⑤ $f(x) = \dfrac{2-x}{x-5}$

$$\lim_{x\to 5^+} \frac{2-x}{x-5} \left[\frac{\neq 0 \text{ (neg)}}{\to \text{kleiner}}\right] = -\infty$$

$$\lim_{x\to 5^-} \frac{2-x}{x-5} \left[\frac{\neq 0 \text{ (neg)}}{\to \text{neg klein}}\right] = +\infty$$

• $D_f = \mathbb{R}\setminus\{5\}$



VA is $x=5$?

Toets

VA is $x=5$

---

⑥ $f(x) = \dfrac{x^2-16}{x-4}$

$D_f = \mathbb{R}\setminus\{4\}$

Is $x=4$ VA ??

$$\lim_{x\to 4} \frac{\cancel{(x-4)}(x+4)}{\cancel{x-4}} \left(\frac{0}{0}\right)$$

onbepaalde vorm

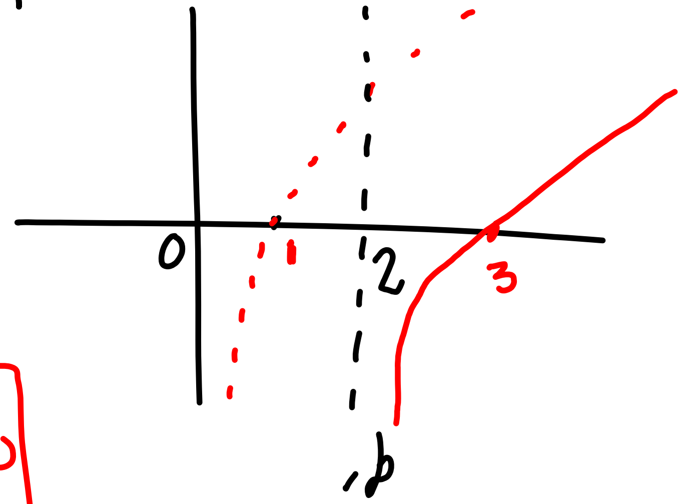
$$= \lim_{x\to 4} x+4 = 8$$

Geen VA

⑦ $f(x) = \ln(x-2)$ $D_f = (2,\infty)$

$$\lim_{x\to 2^+} \ln(x-2) = -\infty$$

Dan $x=2$ VA



• $\ln(x-2) \leq 0$

$\boxed{\begin{array}{l} ① \ \ln(x-2) \leq 0 \\ \quad\quad \text{en} \\ ② \ \ x-2>0 \end{array}}$

$\ln(x-2) \leq 0$

$e^{\ln(x-2)} \leq e^0$

$\Rightarrow x-2 \leq 1 \Rightarrow x \leq 3$ En $x>2$

$\therefore x \in (2,3]$

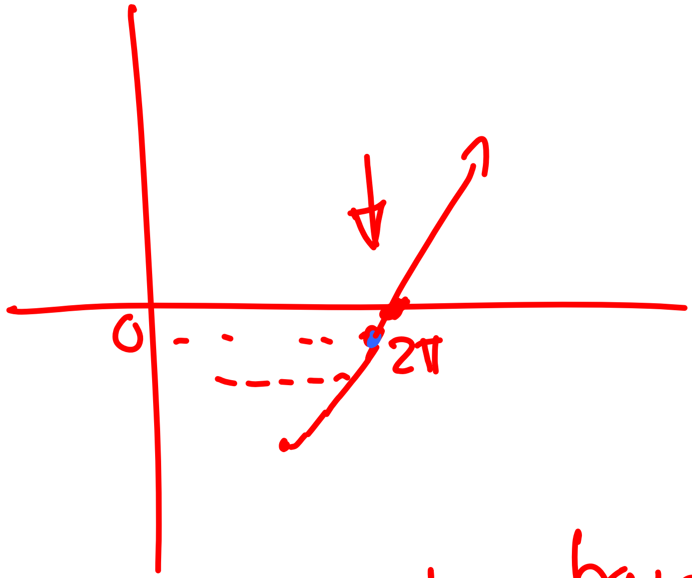
Onthou Tegniekbemeestering agter in Studiegids.

$$\lim_{x\to 2\pi^-} x \csc x$$

$$= \lim_{x\to 2\pi^-} \frac{x}{\sin x} = -\infty$$

$\neq 0$ pos

(neg)



y waardes baie klein neg

$$\lim_{x\to 2\pi^-} x \cot x$$

$$= \lim_{x\to 2\pi^-} \frac{x \cos x}{\sin x} = -\infty$$

2π (1) $\neq 0$

klein neg

of $\lim_{x\to 2\pi^-} \frac{x}{\tan x}$